

Lecture Notes February 24

Orthonormal bases:

Def: let V be inner product space

$S \subseteq V$ is orthogonal if $\forall x, y \in S, x \neq y$,
it holds that $\langle x, y \rangle = 0$

Orthonormal?

if $\langle x, y \rangle = \begin{cases} 1 & \text{if } x=y \\ 0 & \text{otherwise} \end{cases}$

Def: Kronecker delta:

$$\delta_{i,j} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$$

$\{v_1, \dots, v_n\}$ is orthonormal $\Leftrightarrow \langle v_i, v_j \rangle = \delta_{ij}$

Remark: $\forall v \neq 0, \frac{1}{\|v\|} \cdot v$ is a unit vector

Proposition: S orthogonal and $\forall v \in S, v \neq 0 \Rightarrow S$ linearly ind.

Proof: Suppose $\sum_{v \in S} \alpha_v \cdot v = 0$ ($\star$)

Let $u \in S$. Take inner product of $\star$ and u .

$$\alpha_u \cdot \langle u, u \rangle = \sum_{v \in S} \alpha_v \langle u, v \rangle =$$

Prop: V IPS, $B(v_1, \dots, v_n)$ orthonormal basis

Then $\forall x, y \in V, \langle x, y \rangle = \sum_{i=1}^n \langle x, v_i \rangle \overline{\langle y, v_i \rangle}$

$$\Rightarrow \text{If } x=y: \|x\|^2 = \langle x, x \rangle = \sum_{i=1}^n \langle x, v_i \rangle \overline{\langle x, v_i \rangle} = \sum_{i=1}^n \langle x, v_i \rangle^2$$

